

Lakeflow Connect with Supabase Assignment Completion Report

Platform Stack: Supabase Cloud DB | Databricks Lakehouse Architecture

This report documents the end-to-end process followed to create a Supabase database, connect it to Databricks using Lakeflow Connect, configure an Ingestion Pipeline, and validate Change Data Capture (CDC) functionality.

1. Creating Lakeflow Connect

Step 1: Create a Supabase Account

A free Supabase account was created using a GitHub account. After signing in, a personal organization was created and a new Supabase project was provisioned.

Important: During project creation, a database password was configured. This password was saved because it is required later while connecting Supabase to Databricks.

Step 2: Retrieve Database Connection Details

From the Supabase dashboard, the **Connect** option was selected. The **Direct Connection String** method was used, followed by the **Session Pool** configuration. This provided the PostgreSQL connection details such as host name and other connection parameters. These details are later used in Databricks.

Step 3: Create Source Table in Supabase

A table named `orders_practice` was created in Supabase and SQL queries were executed to create and populate the database objects required for the exercise.

Step 4: Create a Lakeflow Connect Connection in Databricks

After logging in to the Databricks workspace, Catalog Explorer was opened and a new connection was created with the following parameters:

- **Connection Type:** PostgreSQL
- **Host and connection details:** Retrieved from Supabase
- **Password:** The password configured during Supabase project creation
- **Test Connection:** Executed using an existing cluster

Once the test completed successfully, the connection was established. The table `orders_practice` became visible under the **public** schema in Catalog Explorer.

2. Creating the Ingestion Pipeline

The next task was to create an Ingestion Pipeline so that data from Supabase could be loaded into the Databricks Lakehouse.

Step 1: Open Jobs and Pipelines

Navigated to the left navigation menu, opened the **Jobs and Pipelines** section, and selected **Create Ingestion Pipeline**.

Step 2: Select Source Type

Searched for **PostgreSQL** as the source type and selected the Lakeflow connection that was created earlier.

Step 3: Configure the Pipeline

Set up the infrastructure and operational rules for the processing system:

- Assign a pipeline name.
- Select an existing catalog for event logs.
- Create a new schema named `raw_schema`.
- Select **Serverless Compute**.
- Click **Create Pipeline and Continue**.

Step 4: Select Source Database

Select the source database from Supabase. Under the **public** schema, choose the table `orders_practice`.

Step 5: Configure Table Settings

Map the individual processing criteria to match tracking rules:

- **Cursor Column:** `updated_at`
- **Primary Key:** `order_id`
- **History Tracking:** Disabled

Step 6: Configure Destination Settings

The destination schema was explicitly set to `raw_schema`.

Step 7: Configure Scheduling and Notifications

The default hourly schedule was removed and failure notifications were selected. The pipeline was then created.

After successful deployment, the table became available within the Databricks catalog under `raw_schema`.

3. CDC (Change Data Capture) in Action

To validate that the ingestion pipeline was working correctly, changes were made to the source table in Supabase.

A new row was inserted into the `orders_practice` table. Once the change was committed, the pipeline was executed again.

To verify that Databricks received the latest updates, the following verification query was executed:

```
SELECT COUNT(*) AS total_orders FROM <your-catalog>.raw_schema.orders_practice;
```

The catalog name can be copied directly from Catalog Explorer and substituted into the query. By comparing the results and validating the newly inserted records, it was confirmed that the ingestion pipeline successfully synchronized the changes from Supabase to Databricks.

Assignment Outcome

The assignment was completed successfully. A Supabase PostgreSQL database was created, connected to Databricks through Lakeflow Connect, an Ingestion Pipeline was configured, and CDC functionality was validated by confirming that new records were synchronized into the Databricks Lakehouse environment.